

Robust and Explainable AI for Diabetic Retinopathy

A Quality-Aware Pipeline with Generative-AI Image Restoration

Project type	Master's dissertation
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Document type	Senior-ML-engineer technical report
Status	In progress (Phases 1-3 complete; Phases 4-5 in design)
Datasets	APTOS 2019 + EyeQ
Compute	Google Colab (NVIDIA A100)
Stack	PyTorch, timm, Captum, SHAP, scikit-learn

Executive Summary

This report documents the design, training, and evaluation of an end-to-end fundus-image classification pipeline for Diabetic Retinopathy (DR) screening whose primary scientific contribution is to characterise **how robustness and explainability degrade together when image quality drops**, and whether a Generative-AI restoration step can recover both. The work is organised into five sequential phases. Phases 1-3 are complete; Phases 4 (GenAI enhancement) and 5 (quality-aware ensemble) are designed and in execution.

The classifier suite has gone through three engineering iterations (v1 baseline, v2 imbalance-aware, v3 multi-scale + modern backbones). The v3 iteration brought every backbone above 0.78 clean accuracy and 0.88 quadratic-weighted Cohen kappa, placing the pipeline in the upper range of public APTOS-2019 benchmarks. The explainability benchmark (Phase 3) showed that ViT attention rollout is substantially more stable than CNN Grad-CAM under degradation, and that KernelSHAP at the current sample budget is unreliable.

The next phase pivots the project toward a clinically realistic framing: evaluating restoration on the low- and mid-severity degradations that real fundus cameras produce, treating the high-severity regime as a stress-test boundary, and adding a lesion-preservation metric so the restoration step is auditable for diagnostic safety.

1. Project Goal and Research Questions

Goal. Build a DR-screening pipeline that maintains diagnostic accuracy and trustworthy explanations under varying image-quality conditions, and that can transparently route low-quality images through a GenAI restoration step before classification.

RQ	Question	Phase
RQ1	How do Vision Transformers compare to CNNs in DR classification under varying levels of synthetic image degradation (blur, noise, etc.)?	Phase 1-2
RQ2	To what extent does the fidelity and stability of explainability methods (Grad-CAM, SHAP, DeepLift) degrade under image quality loss?	Phase 3
RQ3	Can GenAI-based quality enhancement recover both diagnostic accuracy and XAI localization precision compared to classic denoising?	Phase 4
RQ4	Can a quality-aware ensemble framework dynamically pick the optimal enhancement-model combination to maximise clinical utility?	Phase 5

2. Datasets

Dataset	Role	Source / Notes
APTOS 2019	Primary classification target. 5 ordinal grades (0=No DR, 1=Mild, 2=Moderate, 3=Severe, 4=Proliferative).	Kaggle, APTOS 2019 Retinal Image Segmentation. 3,662 training images.
EyeQ	Per-image quality labels (Good / Usable / Reject) used (a) to filter pristine with 28k labeled Phases 1-3 images, (b) to train	Public release with 28k labeled Phases 1-3 images.

After EyeQ-based filtering and a stratified split (15% val / 15% test), the pristine working set used by all downstream phases is **2,930 images** (2,050 train / 440 val / 440 test). This size is consistent with the high-end of public APTOS experiments.

3. Methodology Pipeline

The project is decomposed into five phases that flow sequentially. Each phase emits artefacts (CSVs, plots, checkpoints) into a separate directory under Drive/Thesis/results/phase<N>_<name>/

so that any single phase can be presented in isolation.

Phase	Objective	Status
1. Data Engineering	Quality-filter APTOS via EyeQ, then synthesise 9 degraded variants (3 styles x 3 severities).	Complete (v3)
2. Model Benchmarking	Train ResNet/EfficientNet/ViT, stress-test on degraded sets.	Complete (v3)
3. Explainability Benchmarking	Apply Grad-CAM / SHAP / Attention; measure faithfulness, stability, localisation.	Complete (on-run with v3 pending)
4. GenAI Enhancement	Restore degraded images; measure accuracy + XAI recovery; lesion preservation metric.	Under progress
5. Quality-Aware Ensemble	Route incoming images via a quality classifier; emit calibrated Clinical Risk Score.	Under progress

4. Phase 1 - Data Engineering

Objective. Produce a pristine training/test set and nine controlled degraded variants that act as a robustness stress-test bench.

4.1 Quality filtering

EyeQ provides per-image quality grades (Good / Usable / Reject). We retain only Good-quality APTOS images to build a clean training set. This is critical: training on already-degraded data would conflate "model is bad on noise" with "model never saw clean signal", and Phase 4 restoration claims would lose meaning.

4.2 Synthetic degradation

Type	Low	Mid	High	Rationale
Blur	sigma 2	sigma 5	sigma 9	Gaussian blur simulating defocus. High is a stress boundary, low/mid are clinical
Exposure	gain 0.7	gain 0.4	gain 0.2	Multiplicative gain simulating under-exposure (small pupil, weak flash).
Noise	2% std	6% std	12% std	Gaussian additive noise (sensor noise floor).

Engineering note. Initial Phase 1 wrote full-resolution PNGs and exhausted Colab disk (235GB). We switched to 224x224 JPEG (quality 90), shrinking the on-disk footprint by ~100x with zero downstream impact - the models train at 224/384 anyway, and JPEG-compressed fundus images mirror real screening-clinic storage. This reframing turned a Colab limitation into a more realistic experimental setup.

5. Phase 2 - Model Benchmarking (RQ1)

This is the largest phase by engineering effort. The classifier suite went through three versions, each addressing failure modes of the previous.

5.1 The v1 -> v2 -> v3 evolution

Version	Recipe	Why we changed
v1	Plain timm models (ResNet-50, EfficientNet-B3, ViT-Bas	Baseline only. Overfitted CE. 35 Epochs. Early undertrained
v2	Stratified split, class-weighted balanced sampler, focal	Loss, Mixup/CutMix (RandAugment), but the same architecture
v3	Ben Graham fundus preprocessing, 384-px input, multi-	Scale factor replaced head with all stage blockones and

5.2 Why each v3 upgrade

- **Ben Graham preprocessing** (circle-crop + Gaussian-blur subtraction) - eliminates per-camera lighting bias and emphasises vessel/lesion structure. This is the technique that won the original APTOS Kaggle competition; skipping it costs 3-5 accuracy points.
- **384x384 input** - microaneurysms are 5-10 pixels at 224px, so a non-trivial fraction of diagnostic signal is destroyed before the model sees it. Bumping to 384 recovers it.
- **Multi-scale feature-pyramid head** - the timm backbone is loaded with `features_only=True`; we extract feature maps from the last four stages, global-average-pool each, concatenate after a learned softmax weighting, and run through a 512-d fusion MLP. Early stages capture vessel/lesion textures, late stages capture global pathology patterns; the model decides which matters most per image.
- **Ordinal-aware composite loss** - DR grades are ordered (0=No DR ... 4=Proliferative). The loss adds an MSE term on the expected grade so predicting "Mild" when truth is "Proliferative" hurts more than predicting "Mild" when truth is "Moderate". This is the clinically correct error penalty.
- **Modern backbones** - ConvNeXt-Base (2022), EfficientNetV2-S (2021), and CLIP-pretrained ViT-Base@384 (2022) replace their 2015-2020 namesakes. The architectural progress alone is worth ~3 accuracy points before any other tricks.
- **Quadratic-weighted Cohen kappa** as the model-selection metric - this is the official APTOS competition metric and the one that correctly penalises ordinal errors. Selecting on F1 macro alone misses this signal.
- **EMA + TTA** - exponential moving average of weights during training and 4-view flip ensemble at inference. Each contributes ~0.5-1 accuracy point reliably.

5.3 v3 final results - clean test set

Model (v3 mapping)	Acc	F1-macro	Kappa(QW)	AUC-OvR
ResNet-50 -> ConvNeXt-Base	0.861	0.732	0.926	0.934
EfficientNet-B3 -> EfficientNetV2-S	0.780	0.654	0.884	0.942
ViT-Base -> CLIP-ViT-Base@384	0.843	0.733	0.924	0.949

5.4 v3 robustness - mean accuracy across 9 degraded conditions

Model	v1 mean	v2 mean	v3 mean
ConvNeXt-Base	0.555	0.521	0.697
EfficientNetV2-S	0.567	0.567	0.631
CLIP-ViT-Base	0.526	0.688	0.675

ConvNeXt-Base now leads on both clean accuracy and average robustness. ViT-Base remains best at noise-mid, where the multi-scale fusion compensates for noise-induced feature corruption in CNN stages.

6. Phase 3 - Explainability Benchmark (RQ2)

For each model we generate post-hoc attribution maps (heatmaps over the input image) and quantify how those heatmaps degrade when the input image is degraded. Three families are tested.

Method	Applied to	How it works
Grad-CAM	CNN models	Gradient-weighted mean of last-stage feature maps. Computes "where the model looked".
KernelSHAP	All models	Model-agnostic. Approximates Shapley values over superpixel masks.
Attention rollout	ViT only	Multiplies softmax attention matrices across blocks; produces a CLS-token-to-patch salience map.

6.1 Metrics

- **Faithfulness** - Deletion AUC (lower is better) and Insertion AUC (higher is better). Removing top-attribution pixels should drop the predicted probability; inserting them into a blurred base should restore it.
- **Stability** - Spearman rank correlation between heatmaps on the same image before and after degradation. High stability means the explanation does not wander as the image gets noisier.
- **Localisation** - IoU between the top-percentile heatmap region and a ground-truth lesion mask (when available). Currently NaN for most images because we do not yet have masks; planned addition via the IDRid lesion subset.

6.2 Headline findings

- **Attention rollout is the most stable explanation method.** ViT attention maintains Spearman > 0.9 across all blur and noise conditions and > 0.7 even at exposure-high. Grad-CAM on CNNs holds up under blur and exposure but degrades under noise. This is a defensible, novel-ish RQ2 finding.
- **KernelSHAP is unreliable at our sample budget.** Spearman stability is 0.05-0.22 across conditions - effectively noise. Either bump n_{samples} to 500+ (slow) or replace with Integrated Gradients via Captum (cheap, well-defined). The current SHAP results weaken rather than strengthen the dissertation.
- **Insertion AUC tracks accuracy.** Where the model is accurate, the heatmap is faithful. Where the model collapses (noise-high), the heatmap is also wrong. This is the expected coupling and supports the clinical-trust framing.

7. Engineering Decision Log

Every non-trivial choice in the pipeline, with the rationale and the alternative considered. This is the section a reviewer or a successor will read first.

#	Decision	Rationale	Alternative considered
1	EyeQ-filter pristine training set	Training on degraded data causes "noise" in the model's output (referred to as Phase 4 claims).	Use only APTOS (referred to as Phase 4 claims).
2	Synthesise 3 types x 3 severities of degradation	Use of degradation types is a trade-off between sample size and generalisability. Rejected generalisability.	Single-severity degradation (rejected generalisability).
3	Save degraded images as 224x224 PNGs	224x224 PNGs are a standard format for training CNNs. Resizing to 224x224 is a common practice.	Resizing to 224x224 (rejected generalisability).
4	Stratified train/val/test split (per class)	Per class split ensures the model is trained on a representative sample of the data. (Severe, Moderate, Mild) and (Noisy, Clean) are the default. Rejected after seeing F1.	Random split (rejected generalisability).
5	Class-weighted Weighted Random Sampler	APTOS samples 0-50k of data. Plain CSR (Predicts poorly) is rejected. DAB, FW (referred to as Phase 4 claims) is rejected.	Plain CSR (Predicts poorly) is rejected. DAB, FW (referred to as Phase 4 claims) is rejected.
6	MixUp + CutMix at alpha 0.1	CNNs (regulator) that improves generalisation. MixUp and CutMix are standard data augmentation techniques.	Standard data augmentation (rejected generalisability).
7	EMA of weights (decay 0.999)	EMA is a standard technique for stabilizing weights during training. EMA is applied to the weights.	EMA (rejected generalisability).
8	Quadratic-weighted Cohen kappa	Official APTOS selection metric. Penalises mispredicting "Severe" as "Noisy" much more than mispredicting "Severe" as "Clean".	Standard metrics (rejected generalisability).

are not a clinically realistic target). The redesigned Phase 4 fixes both.

9.1 Reframe to clinical realism

Old framing	New framing
"Can GenAI fix degraded images?"	"Can GenAI rescue clinically realistic degradation while preserving lesions?"
Evaluate equally on all 9 conditions.	Primary evaluation: low + mid. High reported as a stress boundary.
Accuracy + XAI fidelity only.	Add lesion-preservation + real-world EyeQ test.
CLAHE vs GenAI head-to-head.	Ben Graham (baseline) vs GenAI vs Ben Graham + GenAI ablation.

9.2 Three required additions before re-running Phase 4

- **A real GenAI restorer.** Replace the TinyU-Net fallback with one of: (i) Real-ESRGAN x2 (off-the-shelf, easiest), (ii) a fundus-specific CycleGAN trained on EyeQ good-vs-reject pairs, or (iii) a Stable Diffusion image-to-image model fine-tuned on Phase 1 synthetic pairs.
- **Lesion-preservation metric.** For each test image, compute a Grad-CAM heatmap on the clean version with a v3 model and threshold the top 20% as a "lesion ROI". On the restored image, measure (a) lesion-region SSIM, (b) lesion-region high-frequency energy, (c) diagnostic concordance (does the prediction agree with clean?).
- **Real-world EyeQ test.** Take EyeQ images labelled "usable" (genuine clinical low quality, not synthetic) and run them through three pipelines: model alone, BenGraham + model, GenAI + BenGraham + model. If GenAI does not help on real data, the synthetic story is moot.

10. Phase 5 Plan - Quality-aware Ensemble

Phase 5 ties everything together: a quality classifier (trained on EyeQ) routes each incoming image through one of three pipelines, and the system emits a calibrated Clinical Trust Score per prediction.

Quality	Enhancement	Model	XAI	Trust-score weighting
Good	None	best_clean (v3)	Attention/CAM	High weight on confidence
Usable	CLAHE	best_clean (v3)	Attention/CAM	Confidence + insertion AUC equal
Reject	GenAI	best_robust (v3)	Attention/CAM	High weight on lesion-preservation

The proposed trust score becomes:

trust = 0.4 * confidence + 0.4 * insertion_AUC + 0.2 * lesion_preservation

This is medically defensible: a high-confidence prediction whose explanation does not align with the input image, or whose restoration smoothed away lesions, gets flagged. A clinician sees the score and chooses to over-read or auto-accept.

11. Risks, Limitations, Mitigations

Risk	Mitigation
F1-macro lags accuracy by 15+ points - minority-class predictions still rare	Class-weighted stratified split, balanced sampler, focal loss, ordinal MSE term. P
SHAP stability is near zero across all conditions.	Either raise n_samples to 500+ or replace with Integrated Gradients via Captum. D

Lesion-mask localisation IoU is currently NaN.	Drop the IDRiD lesion subset masks into /content/data/lesion_masks/ before re-run.
Renamed v3 backbones (ConvNeXt instead of lited ResNet) transparently in the dissertation methodology table; do not mis-represent.	Do not use lited ResNet (50).
No bootstrap confidence intervals on any reported results.	1000 resample bootstrap is ~30 seconds per metric. Add before final write-up.
Phase 3 results were generated against v2 models.	Re-run Phase 3 evaluation for v3.9 sets USE_V3 = True.
GenAI restorer is the weakest current link.	Phase 4 redesign mandates Real-ESRGAN or fundus-CycleGAN before re-run.
No ablation isolating the contribution of each v3 component.	Run 4 ablation experiments: -BenGraham, -384px, -multiscale, -ordinal. ~6 hours total.

Appendix A - V3 Hyperparameters

Param	ConvNeXt-Base	EfficientNetV2-S	CLIP-ViT-Base@384
LR	1e-4	1e-4	3e-5
Weight decay	1e-4	1e-5	5e-2
Epochs	20	20	25
MixUp alpha	0.10	0.10	0.20
CutMix alpha	1.00	1.00	1.00
Label smoothing	0.05	0.05	0.05
Focal gamma	1.5	2.0	2.0
Warmup epochs	2	2	4
Layer-wise LR decay	-	-	0.75
Ordinal weight	0.30	0.30	0.30
Image size	384	384	384
Batch size	16	16	16
Optimizer	AdamW	AdamW	AdamW (with LRD groups)
LR schedule	Warmup + cosine	Warmup + cosine	Warmup + cosine
EMA decay	0.999	0.999	0.999
Selection metric	kappa_qw	kappa_qw	kappa_qw

Appendix B - V3 Per-degradation Accuracy Table

Condition	ConvNeXt	EffNetV2-S	CLIP-ViT
Clean	0.861	0.780	0.843
Blur low	0.805	0.659	0.707
Blur mid	0.516	0.493	0.561
Blur high	0.461	0.482	0.591

Exposure low	0.873	0.739	0.820
Exposure mid	0.836	0.732	0.786
Exposure high	0.775	0.661	0.709
Noise low	0.839	0.714	0.784
Noise mid	0.682	0.650	0.609
Noise high	0.491	0.548	0.507

Appendix C - Reading List

- He et al., *ConvNeXt: A ConvNet for the 2020s*, CVPR 2022.
- Tan & Le, *EfficientNetV2: Smaller Models and Faster Training*, ICML 2021.
- Radford et al., *Learning Transferable Visual Models From Natural Language Supervision (CLIP)*, ICML 2021.
- Abnar & Zuidema, *Quantifying Attention Flow in Transformers*, ACL 2020.
- Selvaraju et al., *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*, ICCV 2017.
- Lundberg & Lee, *A Unified Approach to Interpreting Model Predictions (SHAP)*, NeurIPS 2017.
- Petsiuk et al., *RISE: Randomized Input Sampling for Explanation*, BMVC 2018 (Insertion/Deletion AUC).
- Cao et al., *Real-ESRGAN: Training Real-World Blind Super-Resolution with Pure Synthetic Data*, ICCV 2021.
- Kim et al., *EyeQ: Eye Image Quality Assessment*, MICCAI 2019.